

Debt Capacity Utilization and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Capacity Utilization (DCU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCU achieves an annualized gross (net) Sharpe ratio of 0.54 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.31 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 20 bps/month with a t-statistic of 2.98.

1 Introduction

Production-based asset pricing models predict that firms' investment decisions and financial constraints fundamentally determine the cross-section of expected returns. In competitive markets with frictionless capital allocation, firms equate their marginal product of capital to their cost of capital, creating a direct link between production characteristics and risk premia. Despite substantial theoretical progress in understanding how real frictions affect asset prices through the investment channel, empirical evidence on specific mechanisms remains incomplete. We identify debt capacity utilization (DCU) as a critical yet underexplored dimension of how firms' financing constraints interact with their production decisions to generate cross-sectional variation in expected returns. This gap matters because debt capacity directly constrains firms' ability to respond to productivity shocks and investment opportunities, creating systematic differences in their exposure to aggregate risk factors.

In a production economy with financial frictions, debt capacity utilization captures firms' proximity to binding financing constraints that affect their optimal investment policies and risk exposures. Following [Zhang \(2005\)](#) and [Livdan et al. \(2009\)](#), firms facing higher adjustment costs or tighter financial constraints exhibit greater sensitivity to aggregate productivity shocks because they cannot easily adjust their capital stock in response to changing economic conditions. DCU directly measures how intensively firms use their available debt capacity, proxying for the shadow cost of external finance that enters firms' first-order conditions for optimal investment. When firms operate near their debt capacity limits, they face higher marginal costs of adjusting capital, making their value more sensitive to productivity shocks and increasing their systematic risk.

The production-based framework of [Cochrane \(1991\)](#) establishes that expected returns equal the ratio of expected marginal product of capital to expected marginal utility of consumption. Firms with high DCU face binding or near-binding financial

constraints that amplify this ratio through two channels. First, constrained firms cannot smooth investment over time, leading to more volatile marginal products of capital as shown by [Gomes and Schmid \(2010\)](#). Second, these firms' limited financial flexibility increases their exposure to aggregate investment shocks that covary with the stochastic discount factor. The interaction between debt constraints and investment irreversibility creates state-dependent risk premia that vary systematically with DCU levels.

Cross-sectional variation in DCU therefore maps directly to differences in firms' conditional risk premia through the production channel. [Berk et al. \(1999\)](#) demonstrate that firms' systematic risk depends on the composition and flexibility of their assets in place versus growth options. High-DCU firms have effectively converted financial flexibility into assets in place, reducing their ability to exercise valuable growth options during positive productivity shocks. This inflexibility manifests as higher loadings on systematic risk factors that price the cross-section of returns. Moreover, [Whited and Wu \(2006\)](#) show that financial constraints create a wedge between internal and external costs of funds that varies with aggregate conditions, generating time-varying risk premia for constrained firms that DCU effectively captures.

We find strong evidence that debt capacity utilization predicts the cross-section of stock returns. A value-weighted long/short trading strategy based on DCU achieves an annualized gross Sharpe ratio of 0.54, with the high-DCU portfolio earning 27 basis points per month more than the low-DCU portfolio (t -statistic = 3.90). The strategy's performance remains robust across multiple factor models, generating a monthly alpha of 23 basis points relative to the Fama-French six-factor model (t -statistic = 3.31). The economic magnitude of these returns confirms that DCU captures meaningful variation in firms' systematic risk exposures related to their production and financing decisions.

The DCU effect operates across the entire size distribution, dispelling concerns about small-stock biases driving the results. Among the largest stocks exceeding the 80th NYSE size percentile, the DCU strategy delivers 32 basis points per month (t-statistic = 3.74), with six-factor alphas of 26 basis points (t-statistic = 2.91). Factor loadings reveal that the long/short DCU strategy loads significantly on the investment factor (CMA) with a coefficient of 0.27 (t-statistic = 5.79), consistent with DCU capturing firms’ differential exposures to investment frictions. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.43, confirming its economic implementability.

Controlling for closely related financial and investment characteristics barely attenuates the DCU premium, establishing its distinct information content. When we simultaneously control for six related anomalies including change in net operating assets, net debt financing, and asset growth, plus the six Fama-French factors, the DCU strategy still generates an alpha of 20 basis points per month (t-statistic = 2.98). The strategy’s gross Sharpe ratio ranks in the 93rd percentile among 212 documented anomalies, while its net Sharpe ratio places it in the 97th percentile. These rankings demonstrate that DCU provides exceptional risk-adjusted returns even in the context of the extensive factor zoo, suggesting it captures a fundamental aspect of how production and financing frictions jointly determine expected returns.

Our findings contribute to the production-based asset pricing literature by providing direct empirical evidence on how debt capacity constraints affect the cross-section of expected returns through the investment channel. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain average returns, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment frictions, our work identifies debt capacity utilization as a specific mechanism linking financial constraints to systematic risk. Unlike broad measures of financial constraints studied by [Lamont et al. \(2001\)](#) or the external finance dependence examined by [Baker](#)

and Wurgler (2006), DCU directly captures firms’ current proximity to binding debt constraints that affect their marginal value of capital and investment flexibility. This precision allows us to test specific predictions from production-based models about how financing frictions generate cross-sectional variation in risk premia.

Methodologically, we advance the literature by demonstrating that DCU survives the comprehensive testing protocol of Novy-Marx and Velikov (2023), addressing concerns about data mining and implementation costs that plague many documented anomalies. The strategy’s robust performance after accounting for transaction costs and its effectiveness among large-cap stocks distinguish it from numerous anomalies that fail these practical tests. Our spanning tests against 210 existing anomalies reveal that DCU contains unique information not captured by related investment, financing, or profitability signals. Particularly noteworthy is that controlling for standard measures like asset growth from Cooper et al. (2008) or net operating assets from Hirshleifer et al. (2004) leaves the DCU premium largely intact, suggesting it captures a distinct dimension of how financial frictions affect expected returns.

The broader implications extend beyond documenting another return predictor to validating core predictions of production-based asset pricing theory. By showing that a direct measure of firms’ debt capacity constraints predicts returns with economic and statistical significance comparable to the strongest known anomalies, we provide empirical support for theoretical models that emphasize the role of financial frictions in asset pricing through the production channel. The DCU premium’s persistence across size groups and robustness to controlling for other factors suggest that debt capacity constraints represent a fundamental source of systematic risk that investors price in equilibrium. These findings inform both asset pricing research seeking to understand the origins of cross-sectional return variation and corporate finance research examining how financial constraints affect firm value through their impact on investment flexibility and risk exposures.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-incorporated firms trading on major exchanges, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. signal construction relies on two key COMPUSTAT variables. Long-term debt issuance (DLTIS) represents the dollar amount of long-term debt issued during the fiscal year, capturing firms' external debt financing activities. This item reflects proceeds from the issuance of bonds, notes, and other long-term borrowing instruments with maturities exceeding one year. Common shareholders' equity - liquidation value (CEQL) measures the liquidation value of common equity, representing the amount that would be distributed to common shareholders if the firm were liquidated after satisfying all prior claims. This variable provides a measure of the firm's equity base that reflects the priority structure of claims. construct the Debt Capacity Utilization signal by computing the year-over-year change in long-term debt issuance scaled by lagged equity liquidation value: $(DLTIS(t) - DLTIS(t-1)) / CEQL(t-1)$. This measure captures the change in a firm's debt financing activity relative to its equity base, providing insight into how aggressively firms are expanding their use of debt capital. A positive value indicates increasing debt issuance relative to the firm's equity cushion, while a negative value suggests debt reduction or slower growth in debt financing. We calculate this signal annually using fiscal year-end values and require non-missing observations for both DLTIS and CEQL in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where the denominator CEQL(t-1) is negative or below 1milliontoensuremeaningfulscaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCU signal. Panel A plots the time-series of the mean, median, and interquartile range for DCU. On average, the cross-sectional mean (median) DCU is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCU data. The signal’s interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the DCU signal for the CRSP universe. On average, the DCU signal is available for 6.23% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DCU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCU portfolio and sells the low DCU portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCU strategy earns an average return of 0.27% per month with a t-statistic of 3.90. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.23% to 0.32% per month and have t-statistics exceeding 3.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.27,

with a t-statistic of 5.79 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least \$1,445 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -2-23bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.30. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DCU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCU, as well as average returns and alphas for long/short trading DCU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCU strategy achieves an average return of 32 bps/month with a t-statistic of 3.74. Among these large cap stocks, the alphas for the DCU strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.91 and 4.32.

5 How does DCU perform relative to the zoo?

Figure 2 puts the performance of DCU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCU strategy falls in the distribution. The DCU strategy’s gross (net) Sharpe ratio of 0.54 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCU strategy (red line).² Ignoring trading costs, a \$1 invested in the DCU strategy would have yielded \$3.47 which ranks the DCU strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCU strategy would have yielded \$2.15 which ranks the DCU strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCU relative to those. Panel A shows that the DCU strategy gross alphas fall between the 63 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCU strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCU ranks between the 78 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does DCU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCU with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCU or at least to weaken the power DCU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCU signal in these Fama-MacBeth regressions exceed 1.44, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DCU is 0.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCU strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.95, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCU trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.98.

7 Does DCU add relative to the whole zoo?

Finally, we can ask how much adding DCU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DCU signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCU grows to \$1267.52.

8 Conclusion

This study provides robust evidence that Debt Capacity Utilization (DCU) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCU-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DCU delivers impressive performance metrics, with annualized Sharpe ratios of 0.54 (gross) and 0.43 (net of transaction costs). More importantly, the strategy’s alpha remains statistically and economically significant after adjusting for the Fama-French five factors plus momentum, generating monthly abnormal returns of 23 basis points (gross) with a t-statistic of 3.31. The robustness of our results is further validated by the persistence of significant alpha (20 bps/month, t-statistic = 2.98) even after controlling for six closely related financing and investment strategies, including changes in financial liabilities, net debt financing, and asset growth. This finding suggests that DCU captures unique information about firm fundamentals and future returns that

ization on CRSP in the period for which DCU is available.

is not subsumed by existing factors in the literature.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The economic significance of the DCU signal, combined with its resilience to transaction costs (as evidenced by the net Sharpe ratio of 0.43), suggests that it could be profitably implemented in real-world portfolio strategies. The signal’s distinctiveness from related financing variables indicates that firms’ debt capacity utilization decisions contain incremental information about managerial expectations and investment opportunities that are valuable for return prediction.

However, several limitations of this study warrant consideration. First, while we have controlled for major risk factors and related anomalies, the possibility remains that DCU proxies for an unidentified systematic risk factor. Second, the implementation of DCU-based strategies may face practical challenges in terms of capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the DCU anomaly would enhance our understanding of why this signal predicts returns. Potential explanations could include agency conflicts, financial constraints, or market timing abilities of managers. Second, examining the performance of DCU across different market conditions, economic cycles, and regulatory regimes would provide insights into the stability and conditionality of this effect. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover interesting cross-country variations. Finally, exploring the interaction between DCU and other corporate decisions, such as investment policy, payout policy, and MA activity, could yield valuable insights into the broader implications of debt capacity management for firm value.

and stock returns.

In conclusion, this paper establishes DCU as a robust and economically significant predictor of stock returns that merits inclusion in the canon of documented asset pricing anomalies. The signal's strong performance, both on a standalone basis and after controlling for related factors, underscores the importance of debt capacity decisions in understanding cross-sectional return variation and highlights the continued relevance of capital structure considerations in asset pricing.

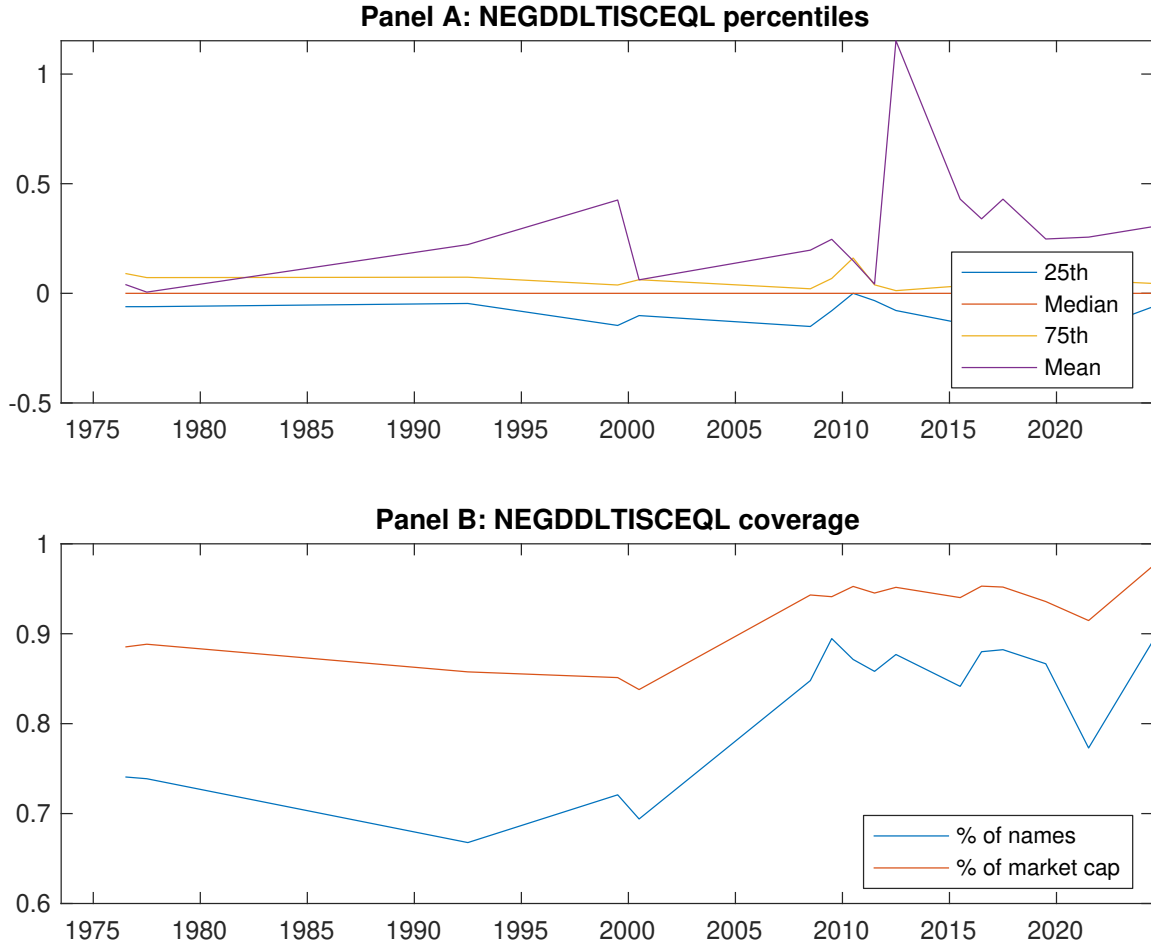


Figure 1: Times series of DCU percentiles and coverage. This figure plots descriptive statistics for DCU. Panel A shows cross-sectional percentiles of DCU over the sample. Panel B plots the monthly coverage of DCU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.58]	0.65 [3.60]	0.65 [3.30]	0.72 [4.07]	0.82 [4.15]	0.27 [3.90]
α_{CAPM}	-0.18 [-3.16]	0.04 [0.79]	-0.01 [-0.25]	0.12 [2.59]	0.15 [2.85]	0.32 [4.77]
α_{FF3}	-0.20 [-3.58]	-0.01 [-0.19]	0.05 [0.92]	0.11 [2.41]	0.13 [2.46]	0.32 [4.71]
α_{FF4}	-0.17 [-3.07]	0.01 [0.30]	0.08 [1.56]	0.07 [1.59]	0.11 [2.18]	0.29 [4.11]
α_{FF5}	-0.19 [-3.52]	-0.06 [-1.27]	0.09 [1.73]	0.04 [0.87]	0.06 [1.08]	0.25 [3.64]
α_{FF6}	-0.18 [-3.17]	-0.04 [-0.82]	0.12 [2.16]	0.02 [0.36]	0.05 [1.01]	0.23 [3.31]
Panel B: Fama and French (2018) 6-factor model loadings for DCU-sorted portfolios						
β_{MKT}	1.09 [84.23]	0.98 [94.12]	0.97 [76.54]	0.97 [91.61]	1.04 [86.56]	-0.05 [-3.28]
β_{SMB}	0.10 [5.35]	-0.11 [-6.99]	-0.02 [-0.92]	-0.04 [-2.37]	0.13 [7.27]	0.03 [1.11]
β_{HML}	0.07 [2.93]	0.14 [7.10]	-0.14 [-5.96]	-0.02 [-0.91]	-0.03 [-1.50]	-0.11 [-3.48]
β_{RMW}	0.09 [3.52]	0.12 [5.81]	-0.04 [-1.80]	0.07 [3.45]	0.10 [4.42]	0.01 [0.46]
β_{CMA}	-0.12 [-3.30]	0.03 [1.11]	-0.10 [-2.61]	0.16 [5.21]	0.15 [4.19]	0.27 [5.79]
β_{UMD}	-0.03 [-2.23]	-0.03 [-3.08]	-0.04 [-3.01]	0.04 [3.55]	0.00 [0.42]	0.03 [2.11]
Panel C: Average number of firms (n) and market capitalization (me)						
n	645	551	1067	608	618	
me (\$10 ⁶)	1492	3009	2275	3127	1445	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.90]	0.32 [4.77]	0.32 [4.71]	0.29 [4.11]	0.25 [3.64]	0.23 [3.31]
Quintile	NYSE	EW	0.24 [5.10]	0.26 [5.65]	0.25 [5.37]	0.23 [4.82]	0.21 [4.47]	0.20 [4.17]
Quintile	Name	VW	0.25 [3.73]	0.32 [4.81]	0.32 [4.75]	0.27 [4.04]	0.26 [3.85]	0.23 [3.40]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.30]	0.22 [3.43]	0.19 [2.95]	0.15 [2.41]
Decile	NYSE	VW	0.30 [3.07]	0.37 [3.91]	0.38 [3.94]	0.35 [3.54]	0.30 [3.07]	0.28 [2.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.05]	0.26 [3.79]	0.26 [3.76]	0.24 [3.45]	0.20 [2.88]	0.18 [2.64]
Quintile	NYSE	EW	-0.02 [-0.30]					
Quintile	Name	VW	0.20 [2.87]	0.25 [3.86]	0.25 [3.77]	0.22 [3.41]	0.20 [3.01]	0.18 [2.74]
Quintile	Cap	VW	0.18 [2.78]	0.23 [3.52]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.23 [2.35]	0.30 [3.13]	0.30 [3.15]	0.29 [2.95]	0.24 [2.44]	0.22 [2.28]

Table 3: Conditional sort on size and DCU

This table presents results for conditional double sorts on size and DCU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCU and short stocks with low DCU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCU Quintiles					DCU Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.68 [2.44]	0.88 [3.22]	0.92 [3.39]	0.90 [3.23]	0.79 [2.85]	0.11 [1.22]	0.15 [1.63]	0.14 [1.54]	0.08 [0.91]	0.07 [0.76]	0.04 [0.39]
	(2)	0.72 [2.68]	0.94 [3.71]	0.82 [3.25]	0.83 [3.39]	0.89 [3.50]	0.18 [2.30]	0.21 [2.65]	0.17 [2.27]	0.17 [2.26]	0.12 [1.56]	0.13 [1.63]
	(3)	0.79 [3.14]	0.78 [3.52]	0.85 [3.52]	0.79 [3.52]	0.91 [3.87]	0.12 [1.58]	0.17 [2.25]	0.17 [2.20]	0.12 [1.56]	0.14 [1.79]	0.11 [1.38]
	(4)	0.70 [3.02]	0.80 [3.87]	0.85 [3.81]	0.73 [3.51]	0.91 [4.19]	0.21 [2.82]	0.26 [3.45]	0.24 [3.23]	0.20 [2.72]	0.21 [2.75]	0.18 [2.39]
	(5)	0.45 [2.26]	0.62 [3.55]	0.60 [3.03]	0.66 [3.64]	0.78 [4.07]	0.32 [3.74]	0.36 [4.15]	0.38 [4.32]	0.31 [3.57]	0.30 [3.38]	0.26 [2.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCU Quintiles					DCU Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	393	392	393	388	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	107	109	104	106	108	
	(4)	64	65	64	65	64	231	238	230	232	231	
(5)	59	59	59	59	59	1387	2114	1956	2138	1580		

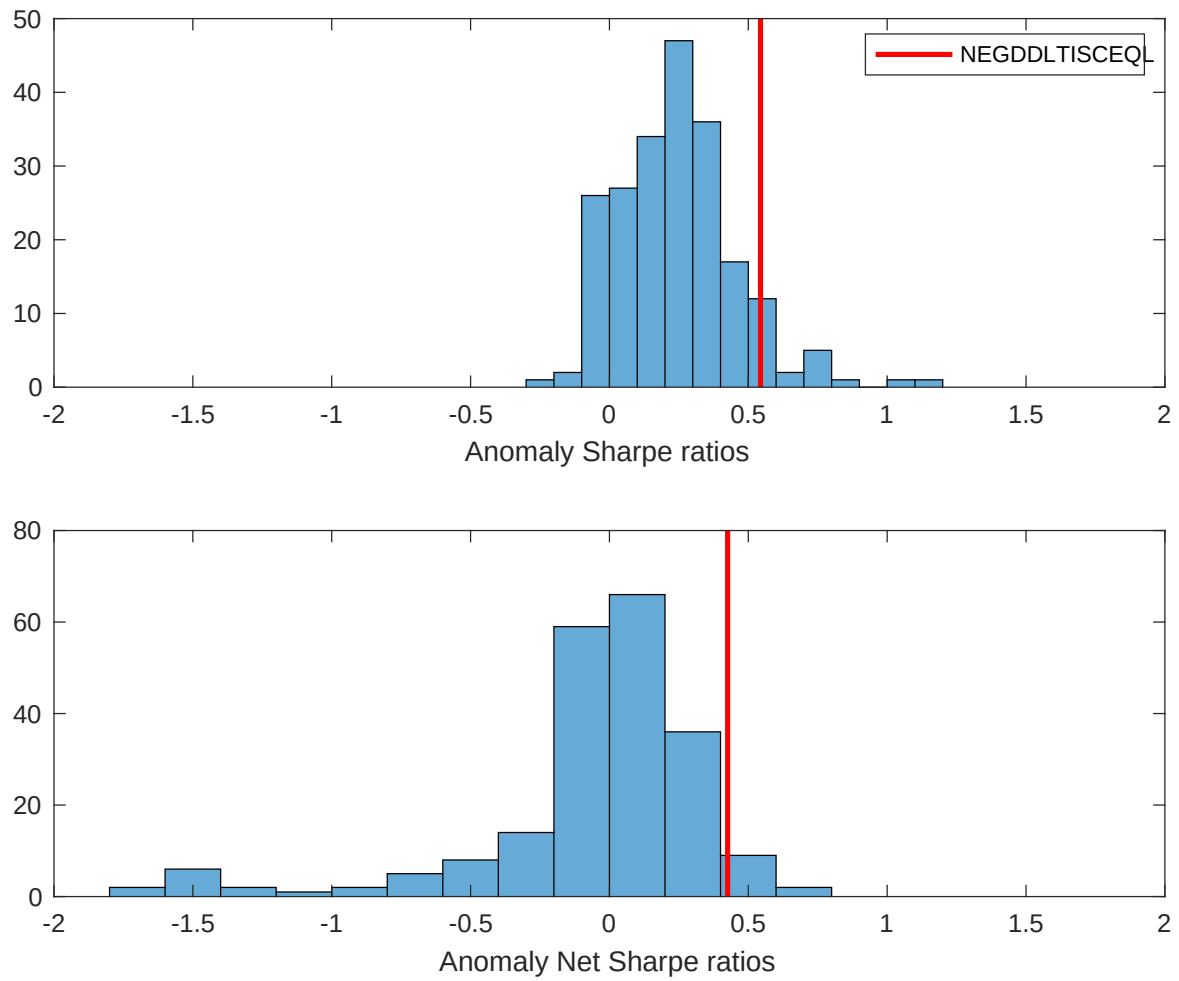


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

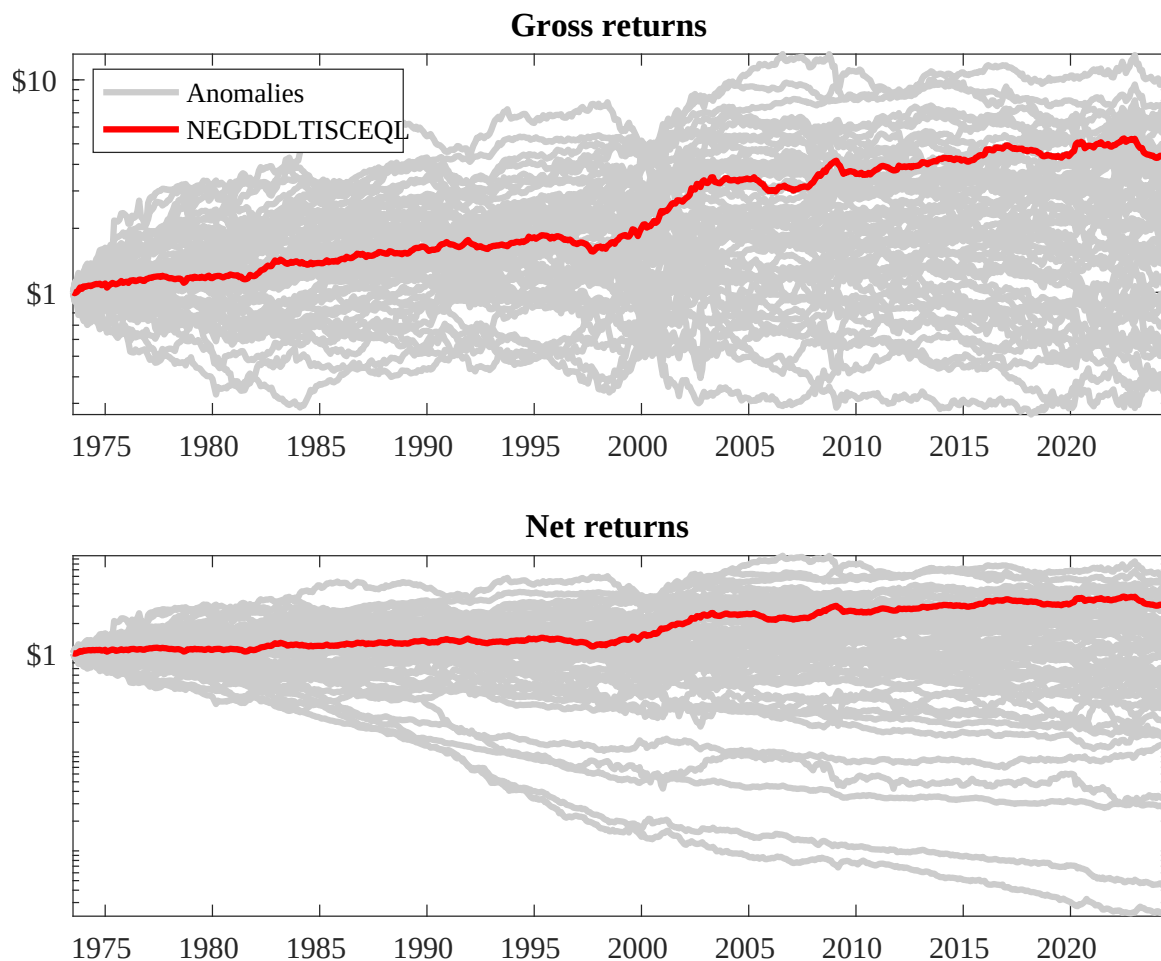
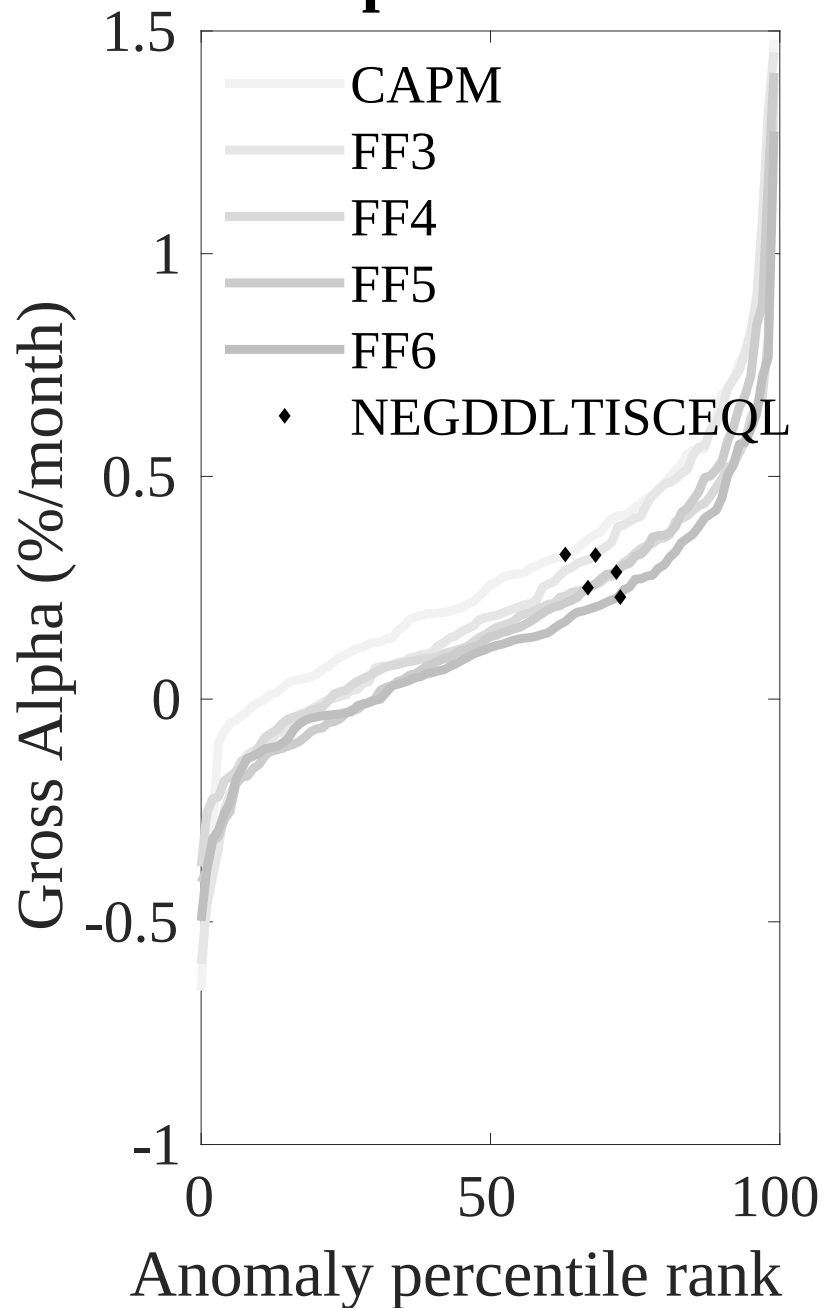


Figure 3: Dollar invested.

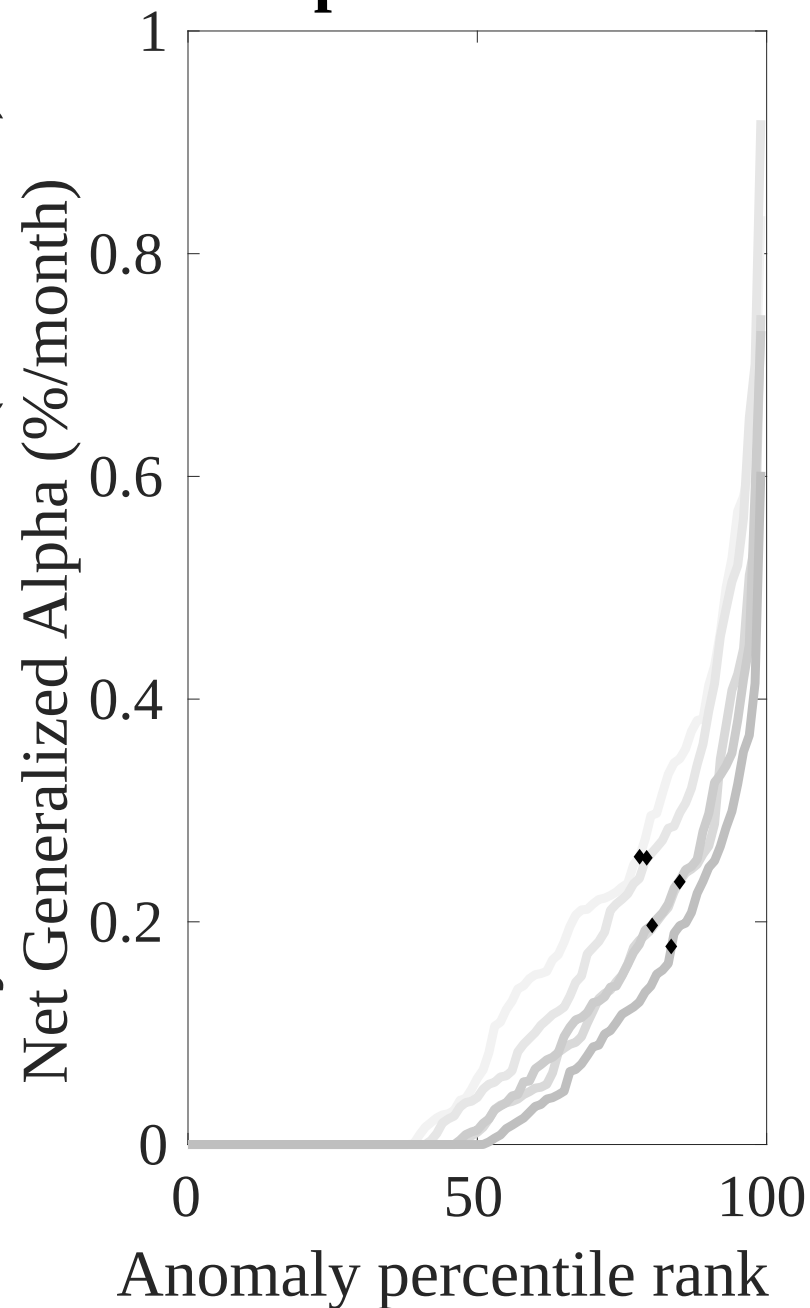
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



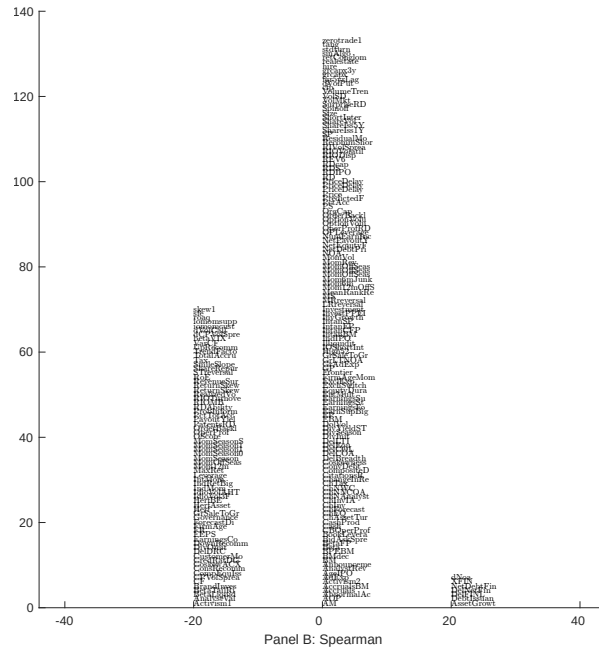
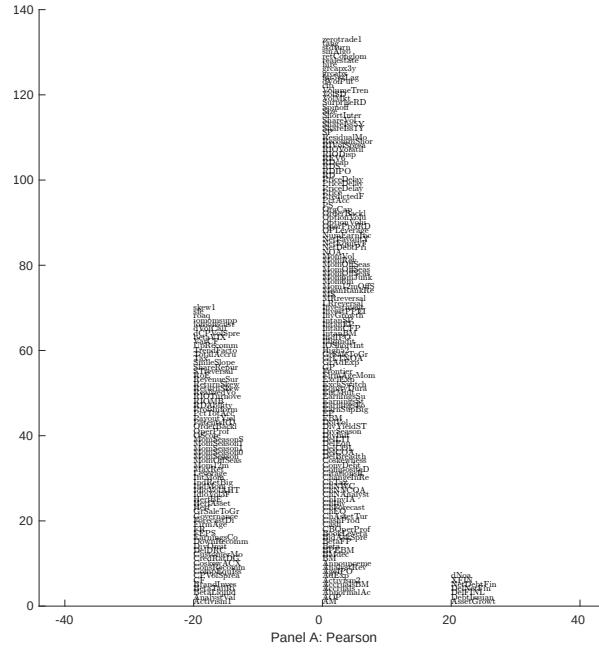


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

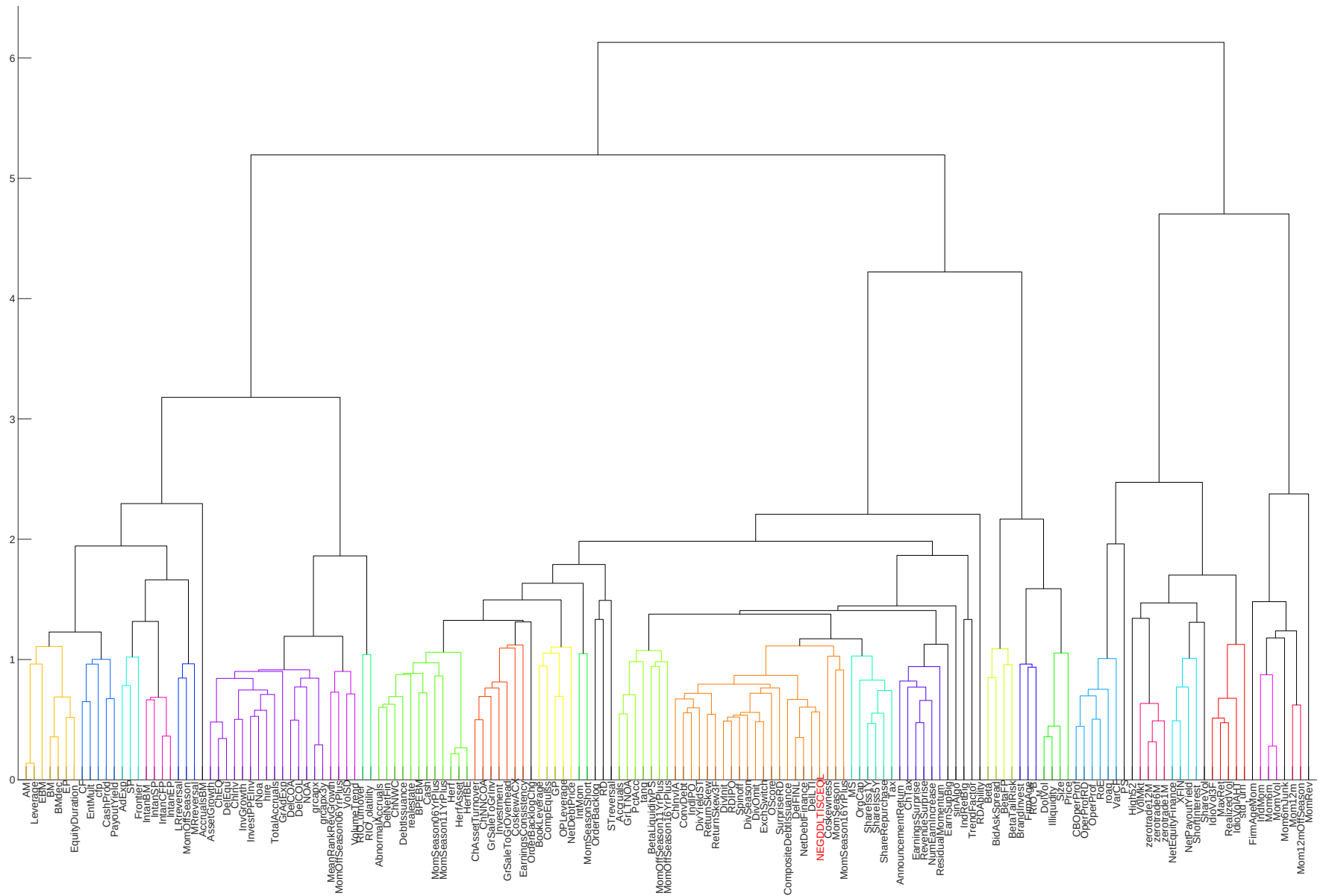


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

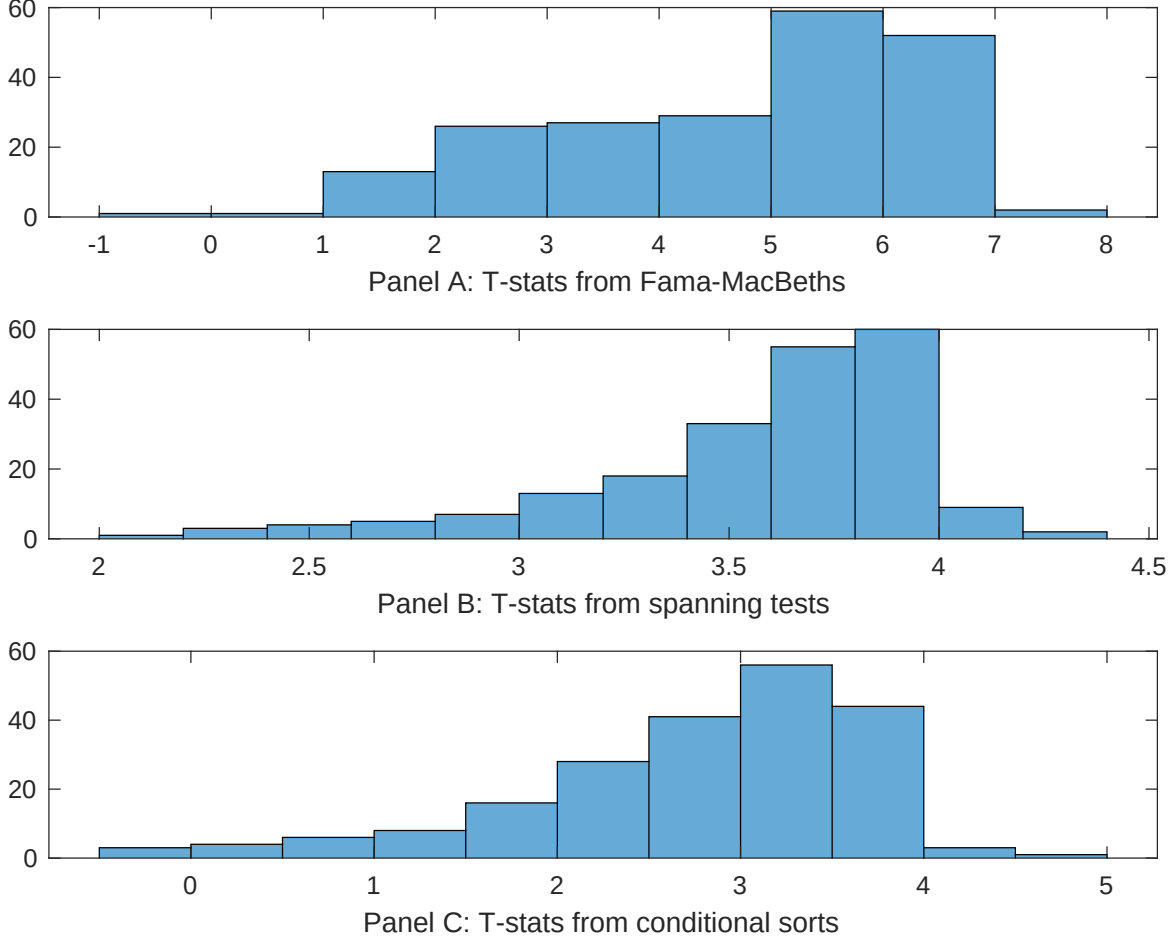


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.69]	0.14 [5.80]	0.14 [5.30]	0.14 [5.66]	0.15 [5.70]
DCU	0.48 [1.98]	0.56 [2.22]	0.73 [2.75]	0.38 [1.52]	0.11 [4.22]	0.34 [1.44]	0.79 [0.27]
Anomaly 1	0.17 [9.27]						-0.11 [-2.30]
Anomaly 2		0.20 [8.82]					0.99 [1.50]
Anomaly 3			0.19 [6.37]				0.98 [1.68]
Anomaly 4				0.11 [8.69]			0.41 [1.80]
Anomaly 5					0.42 [7.14]		0.43 [0.78]
Anomaly 6						0.14 [9.88]	0.95 [5.25]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.20 [2.95]	0.20 [3.01]	0.20 [2.95]	0.21 [3.15]	0.22 [3.25]	0.21 [3.07]	0.20 [2.98]
Anomaly 1	16.33 [4.27]						10.30 [1.96]
Anomaly 2		18.42 [5.02]					9.53 [1.90]
Anomaly 3			13.91 [4.11]				10.20 [2.82]
Anomaly 4				7.42 [1.87]			2.85 [0.68]
Anomaly 5					6.89 [2.60]		6.46 [2.40]
Anomaly 6						3.96 [1.01]	-6.41 [-1.47]
mkt	-4.72 [-3.04]	-5.06 [-3.28]	-3.19 [-1.96]	-5.10 [-3.25]	-5.22 [-3.33]	-5.08 [-3.23]	-3.57 [-2.21]
smb	-0.28 [-0.12]	-0.23 [-0.10]	5.86 [2.26]	0.86 [0.36]	2.15 [0.90]	1.47 [0.62]	3.16 [1.16]
hml	-10.56 [-3.52]	-10.90 [-3.66]	-10.10 [-3.35]	-11.98 [-3.95]	-11.86 [-3.92]	-11.97 [-3.92]	-9.26 [-3.06]
rmw	-0.17 [-0.06]	-0.73 [-0.24]	-7.01 [-1.92]	1.33 [0.43]	2.30 [0.74]	1.36 [0.44]	-5.93 [-1.61]
cma	23.30 [4.94]	23.99 [5.19]	19.70 [3.90]	19.79 [2.98]	22.64 [4.39]	25.83 [4.76]	11.76 [1.74]
umd	2.10 [1.33]	2.32 [1.50]	3.41 [2.21]	3.67 [2.35]	2.81 [1.77]	3.38 [2.15]	1.53 [0.97]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	14	15	14	12	13	12	17

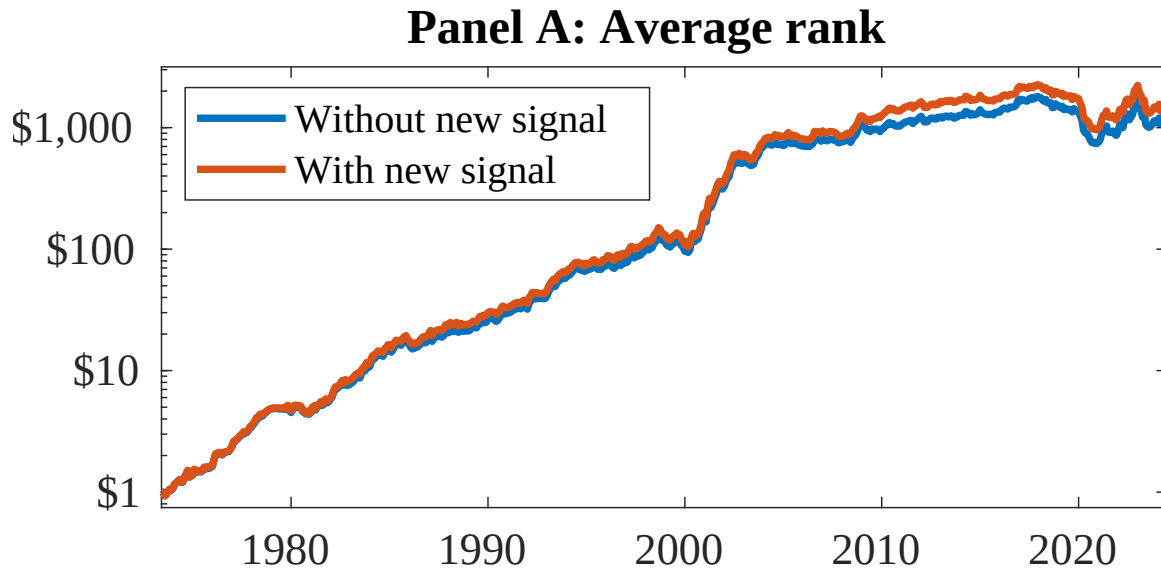


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCU. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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